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Sense of agency promotes recognition memory: the role of self-referential processing

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ABSTRACT

Recent research has explored whether sense of agency (SoA) over to-be-remembered items enhances performance in recognition memory tasks, and what the underlying mechanisms might be. The present study systematically investigates the role of self-referential processing in this improvement. Building on the Comparator Model of SoA, we manipulated SoA from two perspectives: retrospective inference (action-outcome consistency) in Experiment 1 and 2, and prospective inference (action volition) in Experiment 3 and 4. In experiment 1 and 3, we replicated the effect of SoA on recognition memory with a simple recognition task. The results showed that stronger induced SoA significantly improved recognition memory performance. Experiment 3 and 4 used a Remember/Know (R/K) memory paradigm, measured recollection/familiarity performance, and recorded Event-related potentials (ERP). SoA influenced R/K performance in Experiment 4, as evidenced by both behavioral and ERP results, specifically the R performance, FN400 and LPC components were all higher with volition induced SoA. In contrast, Experiment 2 did not reveal a significant R/K improvement effect. These findings suggest that voluntary actions, which carry more self-referential information as compared to involuntary actions, facilitate the SoA and enhance recognition memory. However, the self-referential information associated with action-outcome consistency in Experiment 2 might not have been sufficiently detectable, and its effect may have disappeared when recognition memory was measured using the R/K paradigm, which likely requires more cognitive resources.

1. Introduction

The sense of agency (SoA), which refers to the feeling of controlling one's actions, their outcomes, and the external environment (Gallagher, 2000), plays a fundamental role in various aspects of human life (Haggard, 2017; Sugimori and Asai, 2015). For instance, SoA is closely linked to mental health, where an accurate and effective assessment of SoA can facilitate the prediction and intervention of mental health issues. Abnormal SoA is strongly associated with several mental disorders and is considered as a risk factor for psychiatric conditions (Frith et al., 2000). In the legal context, establishing criminal responsibility requires not only confirming the actions but also evaluating an individual's SoA over those actions (Frith, 2014). Moreover, as technology advances, ensuring that users experience a higher SoA in human-computer interactions becomes increasingly important (Cornelio et al., 2022).

Researchers have proposed models such as the comparator model and the multifactorial two-step account of SoA (Frith et al., 2000; Moore, 2016; Haggard, 2017; Synofzik et al., 2008). These models suggest that a specific action begins with an intention or volition, followed by the generation of a motor command that drives the action to achieve a goal. An efference copy of the motor command is then used to predict the potential consequences of the action. This prediction is compared with the sensory feedback received during the action and its effects. When the prediction aligns with the sensory feedback, a strong feeling of SoA is experienced. If the action's outcome also aligns with the original intention, a strong attribution of SoA is formed (Synofzik et al., 2008).

SoA can be measured directly or indirectly. A direct, explicit measure involves asking participants questions such as, “Did you perform the action and cause the outcome?” This type of judgment requires individuals to attribute sensory events to their own intentional actions. An indirect, implicit measure is the so-called intentional binding effect (Haggard et al., 2002). For example, Haggard et al. (2002) found significant differences in SoA between voluntary actions and actions

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induced by transcranial magnetic stimulation (TMS) of the motor cortex. Participants performing voluntary actions reported a perceptual binding between their actions and outcomes, a temporal compression known as the intentional binding effect. Ebert and Wegner (2010) demonstrated that consistency between one's action and a subsequent event enhanced both the intentional binding effect and self-reported SoA.

Interestingly, the extent to which varying degrees of the SoA influence cognitive functions and behavior remains insufficiently studied. In recent years, several studies have begun to examine the impact of SoA on memory, using different experimental manipulations and theoretical perspectives (Hon and Yeo, 2021; Houser et al., 2022; Tsuji and Imaizumi, 2022; Zou et al., 2023). For example, Hon and Yeo (2021) asked participants to press keys to control the movement of word stimuli during the learning phase and found that increasing action–outcome consistency enhanced memory performance as participants experienced a strong SoA. They attributed this memory enhancement to self-referential processing: when action–outcome consistency was high, participants experienced a stronger SoA, which rendered the outcomes—and the associated word stimuli—more self-relevant. This increased self-relevance was assumed to make the memories more distinctive and to provide additional retrieval cues in the recognition. Tsuji and Imaizumi (2022) later examined this assumption by using the remember/know (R/K) paradigm to investigate whether self-referential processing mediates the effect of SoA on memory. The classic R/K paradigm (Gardiner, 1988), in which participants first make an old/new recognition judgment and, for items judged as old, further indicate whether their recognition is accompanied by recollecting specific contextual details ("Remember") or based on a mere feeling of familiarity ("Know"), thus provides a behavioral distinction between these two subjective recognition experiences: "Remember" responses indicate recollection, while "Know" responses indicate familiarity. Also, previous researches predict that self-awareness and self-referential processing enables better recognition performance of self-related details, so self-relevant information is typically reflected in the recollection component (the "Remember" performance) rather than familiarity (Conway and Dewhurst, 1995). Following this assumption, Tsuji and Imaizumi (2022) assumed that the R response would be significantly higher in consistent than inconsistent condition, while the K response would not. Their results, however, showed that action–outcome consistency did not affect either recollection ("Remember") or familiarity ("Know") responses. These findings leave it unclear whether SoA influences memory within the R/K framework, and cast doubt on the proposed role of self-referential processing.

In the meanwhile, we note that the theoretical interpretation of these R/K responses remains a matter of ongoing debate. As discussed above, under the dual-process framework, Remember and Know responses are assumed to reflect two qualitatively distinct memory processes: recollection and familiarity, respectively (Addante et al., 2012). Within this framework, Remember responses are often associated with the late positive component (LPC; 500–800 ms) over parietal regions, while Know responses are linked to the fronto-central FN400 component (300–500 ms; Curran, 2000; Rugg and Curran, 2007). This framework has been widely adopted in studies investigating the neural correlates of recognition memory, and in Tsuji and Imaizumi (2022). An alternative account, however, interprets R/K judgments within a single-process/signal detection theory (SDT) framework, in which Remember and Know responses reflect different decision criteria applied along a single continuous dimension of memory strength (Dunn, 2004, 2008; Wixted and Mickes, 2010). In this view, a strong memory signal exceeds a high criterion, yielding a Remember response; whereas a weaker signal only exceeds a lower criterion, yielding a Know response. Critically, the same R/K behavioral data can be accommodated by both frameworks, but they lead to different predictions regarding the neural correlates of recognition. In the present study, we adopted the R/K paradigm as a behavioral tool to dissociate different recognition experiences, while remaining open to both theoretical interpretations. In the General Discussion, we revisit these competing accounts in light of our behavioral and Event-related potentials (ERP) findings.

Taken in sum, after careful consideration of these theories and findings, we propose three possible explanations for the discrepancy between the two closely related studies by Hon and Yeo (2021) and Tsuji and Imaizumi (2022).

First, it is possible that SoA induced by action–outcome consistency does not influence recognition memory. However, this explanation appears unlikely, given converging evidence from multiple studies demonstrating that SoA affects memory across a range of manipulation methods (Hon and Yeo, 2021; Houser et al., 2022; Murty et al., 2015; Zou et al., 2023). For instance, beyond direct manipulations of action–outcome consistency (Hon and Yeo, 2021; Tsuji and Imaizumi, 2022), Zou et al. (2023) manipulated SoA by varying the probability of keypress–outcome correspondence and likewise observed significant improvements in recognition memory.

Second, although self-referential processing has been shown to enhance both encoding and retrieval of word stimuli (Caudek, 2014), one might argue that SoA influences recognition memory through mechanisms that are independent of self-referential processing. Under this account, SoA would enhance memory performance, but this enhancement would not manifest as differences in recollection or familiarity, explaining why the R/K paradigm failed to detect effects in the study by Tsuji and Imaizumi (2022). However, evidence from other lines of research suggests that SoA may indeed influence memory via self-referential mechanisms. SoA has been shown to be closely related to self-awareness (Knoblich and Flach, 2003; Sugimori and Asai, 2015), and information related to an individual's self-concept is known to be more deeply processed and better remembered (Rogers et al., 1977). From this perspective, words encoded under conditions of high SoA are more likely to be self-referenced, which could enhance memory performance, the non-self-referential account cannot fully explain the existing evidence.

Accordingly, the present study aimed to further investigate the influence of SoA on recognition memory and to examine its underlying mechanisms from the perspective of self-referential processing. To test the first possible explanation, the Experiment 1 employed a standard recognition memory task (Hon and Yeo, 2021) to replicate the finding that perceived SoA over word stimuli—manipulated solely via action–outcome consistency—enhances recognition memory. And to test the second possible explanation, in Experiment 2, SoA was again manipulated through action–outcome consistency, and the R/K paradigm was used to assess the potential contribution of self-referential processing (Tsuji and Imaizumi, 2022), we also employed the Electrophysiological measures, to differentiate the recollection (R response) and familiarity (K response) components. As the self-referential processing enhances encoding-stage processing and leaves measurable neural signatures at retrieval (e.g., the old/new effect). Especially, within the dual-process framework, the FN400 component, a stimulus-evoked negative wave, is associated with familiarity ("Know"); whereas the LPC component, a stimulus-evoked positive wave, reflects recollection ("Remember") (Curran, 2000; Tsivilis et al., 2015; Addante et al., 2012; Migo et al., 2012). For example, Woodruff et al. (2006) asked participants to rate items at four levels of familiarity within a self-referential R/K paradigm. The results indicated that FN400 amplitude decreased with lower familiarity levels, while LPC amplitude increased for items requiring recollection but did not vary across familiarity levels when recollection was not required. More directly, self-relevant memories have been shown to produce greater LPC positivity (Rugg and Curran, 2007). Thus through the combination of behavioral and ERP data, we might be able to conduct a comprehensive investigation into the possible self-referential effect.

Third, it is possible that SoA does influence recognition memory and that this effect is indeed related to self-referential processing, but that the effect associated with action–outcome consistency is relatively weak. This interpretation is plausible, as prior research has suggested

that SoA may play only a minor role in attentional capture when compared to strong exogenous cues (Vilanova-Goldstein et al., 2022). Consequently, the memory-enhancing effect of SoA may be easily overshadowed by other task-related factors. One such factor is the use of the R/K paradigm in the study by Tsuji and Imaizumi (2022). This paradigm requires participants not only to recognize previously presented words but also to engage in additional evaluative processing to classify their responses as based on recollection ("remember") or familiarity ("know"). This added cognitive demand may mask relatively subtle improvements in recognition memory associated with SoA, such as those reported by Hon and Yeo (2021). Compared to a simple recognition task, the R/K paradigm is therefore more cognitively demanding, and this increased demand may obscure weak SoA-related effects.

Accordingly, in Experiment 3 and 4, the R/K paradigm was retained, but the manipulation of SoA was shifted from SoA feeling experience (action–outcome consistency) to SoA attribution (i.e., whether the action was voluntary or externally imposed on the participant), a manipulation shown in prior research to induce a stronger sense of agency (Houser et al., 2022; Murty et al., 2015). This design allowed us to test whether a more pronounced SoA can overcome the increased cognitive demands of the R/K paradigm and produce measurable improvements in recognition memory. We considered that, in Experiment 3 and 4, enhanced SoA will facilitate memory performance through self-referential processing mechanisms. According to the self-referential processing, the R/K performance would be like this: the R response would be significantly higher in voluntary than involuntary condition, while the K response would not. At the neural level, we expect that the voluntary condition will elicit a larger LPC old/new effect (reflecting recollection), while the FN400 old/new effect (reflecting familiarity) remains unchanged.

2. Experiment 1

In Experiment 1, Tsuji and Imaizumi (2022) did not observe an effect of SoA on recognition memory, a finding that appears to contradict the results reported by Hon and Yeo (2021), who employed a standard recognition memory task. Notably, compared with Hon and Yeo (2021), Tsuji and Imaizumi (2022) did not include a time-delay manipulation of SoA and instead relied on the R/K paradigm to examine the potential role of self-referential processing. These methodological differences may account for the discrepant findings across studies. Accordingly, the Experiment 1 in the present research aimed to replicate the effect of SoA on recognition memory by manipulating action–outcome consistency alone, thereby providing a foundation for subsequent investigations.

2.1. Method

2.1.1. Participants

We performed an a-priori power analysis with G power (Faul et al., 2007), with a medium effect size (f = 0.25), α = 0.05, power = 0.8, number of groups = 1, number of measurements = 2, and found that the required sample size is 34. We then enrolled 34 participants (female 23), all undergraduate students from one university, age range 17–22, mean age 19.71, SD = 1.49, and recruited through online advertisements. Informed consent was obtained from all participants prior to the experiment, and they were compensated for their participation after the study. Experiments were conducted in accordance with the Declaration of Helsinki and approved by the local Ethics Committee of University (approval number H25040).

2.1.2. Experimental materials

A total of 108 neutral two-character Chinese words was randomly selected from the Chinese Affective Words System (Wang et al., 2008), with emotional valence ratings of 5.01 ± 1.50 and arousal level ratings of 4.25 ± 1.81 in the range of 1–9. Among these, 72 words were used for learning, and 36 were new words for testing.

2.1.3. Design

The experiment had a single-factor within-participants design for both SoA and memory task. The independent variable was the action-outcome consistency (consistent vs. inconsistent), and the dependent variable was participants' recognition performance. Explicit SoA was rated in each trial, as a manipulation check (Hon and Yeo, 2021; Tsuji and Imaizumi, 2022).

2.1.4. Procedure

Participants came to the lab and sat in front of a computer screen. The experiment comprised two phases: learning and test. During the learning phase, participants were first presented with a fixation point at the center of the screen (700 ms), which was then replaced by a word. The experimenter indicated to the participants that they could control the word's movement freely by pressing the "↑" or "↓" keys, with a single and rapid press only, and then they could see the word's movement immediately. In consistent conditions, the word moved in the same direction as the pressed key for 400 ms, and then stopped above or below the fixation point; in inconsistent conditions, the word moved into the opposite direction. Subsequently, the word remained visible at its final position for an additional 500 ms.

Participants were asked to memorize the words. There were 36 words to be learned under each condition. Upon completion of learning each word, participants were required to rate their SoA, i.e., "To what extent do you feel it is your key press which caused the movement of the word", using a Likert 7-point scale. Participants were instructed to choose and input a score ranging from 1 to 7, where 1 indicated "not at all", 4 indicated "Not sure" and 7 indicated "very strong". The consistent and inconsistent trials were randomly intermixed during the learning phase.

In the testing stage, half of the words learned under each condition (18 × 2) were randomly selected and mixed with 36 new words, resulting in a total of 72 test words. Participants were exposed to one word in each trial, and they needed to press "F" for words they thought they had learned; and "J" for new words. The detailed process is illustrated in Fig. 1.

2.2. Results

We used both Classical and Bayesian ANOVA analysis, and *p*-values and Bayes Factors were reported complementarily. We interpreted Bayes Factors following Lee and Wagenmakers (2013): anecdotal (BF10 < 3), moderate (3 ≤ BF10 < 10), strong (10 ≤ BF10 < 30), very strong (30 ≤ BF10 ⟨100), and extreme evidence (BF10 ≥ 100). In case of inconsistent outcomes, Bayes-factor-based evidence guided our main inference, while all statistics results were fully reported. Significant *p*-values accompanied by only anecdotal Bayes-factor evidence were interpreted as inconclusive for the hypothesized effect.

We used one factor repeated measure ANOVA to analyze the SoA data, with the within participant factor consistency (consistent or inconsistent). Results indicated a significant main effect of consistency, $F(1,33) = 180.98$, $p < .001$, $\eta p2 = 0.85$, BF10 > 100, providing very strong evidence. The SoA scores were significantly higher in the consistent condition (Mean = 5.91, SE = 0.17) than in the inconsistent condition (Mean = 2.25, SE = 0.20).

We calculated the memory judgments for old and new words in each consistency condition for each participant. The memory performance was calculated by subtracting the false alarm proportion for new words from hit proportion for old learned words for all conditions. In the recognition memory task, with one factor repeated measure ANOVA, as well as the Bayesian analysis, results indicated a significant main effect of consistency, $F(1,33) = 6.94$. $p = .013$, $\eta p2 = 0.17$, and showing that recognition performance scores significantly higher in the consistent condition (Mean = 0.37, SE = 0.03) than in the inconsistent condition

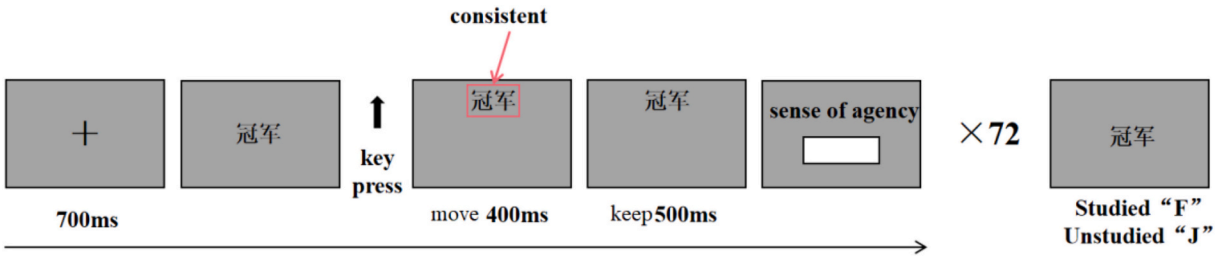


Fig. 1. Experimental procedure in Experiment 1.

(Mean = 0.34, SE = 0.03). Bayesian analysis revealed that BF10 = 3.52, providing moderate evidence for our hypothesis that recognition performance differed between the consistent and inconsistent conditions. Please see Fig. 2.

2.3. Discussion

The Experiment 1 aimed to replicate the finding that SoA, manipulated through action-outcome consistency, can improve recognition memory (Hon and Yeo, 2021; Tsuji and Imaizumi, 2022; Houser et al., 2022; Murty et al., 2015; Zou et al., 2023). The SoA ratings demonstrated that we successfully manipulated the effect of action-outcome consistency on the SoA: Participants experienced a stronger SoA in the consistent condition than in the inconsistent condition, like in previous research (Ebert and Wegner, 2010; Hon and Yeo, 2021). Importantly, recognition memory was significantly higher in the consistent condition than in the inconsistent condition, indicating that a stronger SoA can indeed enhance memory. This finding not only corroborates Hon and Yeo (2021) results but also suggests that the null-effect reported by Tsuji and Imaizumi (2022) might be related to their use of a different paradigm (R/K). This observation set the stage for the next experiments, where we tested the role of the research paradigm and its self-referential implications for the memory enhancement effect.

One might argue that the inconsistent condition, by violating action-outcome predictions, could capture greater attention (Iwanaga and Nittono, 2010; Steinhauser et al., 2018) and thus potentially enhance memory for those items. If this were the case, we would expect better memory in the inconsistent condition, however, which contradicts our observed pattern. Therefore, although prediction errors may indeed capture attention, this attentional effect appears to be insufficient to override the memory-enhancing effect of SoA—or alternatively, the captured attention may be directed primarily to the sensorimotor discrepancy itself, rather than to the identity of the word stimulus. Thus, the present results suggest that the SoA-related encoding advantage dominates over any opposing attentional influence under our experimental settings.

3. Experiment 2

In Experiment 2, consistent with previous research (Tsuji and Imaizumi, 2022), we employed the R/K paradigm to record "Remember" and "Know" recognition performance. To provide more information, we also recorded event-related potentials (ERPs) while participants performed the R/K recognition task (Rotello et al., 2005; Leynes and Mok, 2017; Leynes and Crawford, 2018).

Taken together, in Experiment 2 we continued to investigate the mechanisms by which SoA enhances recognition memory from the perspective of self-referential processing. We included two dependent measures (R/K performance and ERP components). Based on our hypothesis, if SoA improves memory via self-referential processing, we would expect to observe differences in both "Remember" performance and LPC amplitude, with values in the consistent condition be significantly higher than those in the inconsistent and new-word conditions; while expect no difference in either the "Know" performance and FN400 amplitude.

3.1. Method

3.1.1. Participants

We performed an a-priori power analysis with G power (Faul et al., 2007), with a medium effect size (f = 0.25), α = 0.05, power = 0.8, number of groups = 1, number of measurements = 4, and found that the

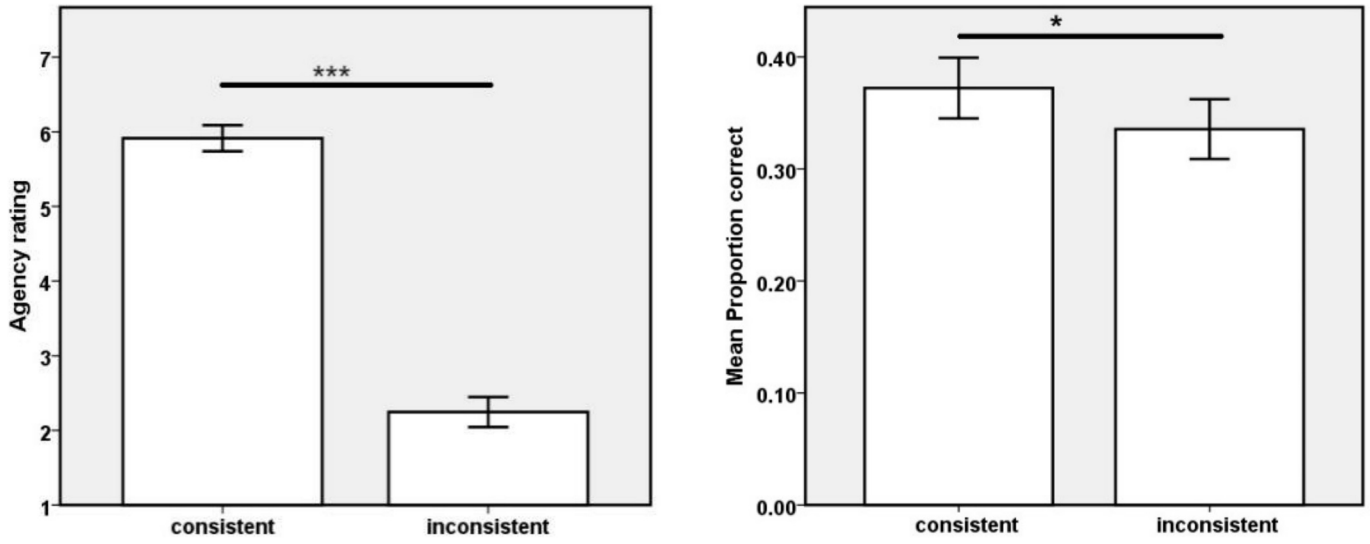


Fig. 2. Agency ratings (left) and Recognition performance (right) in Experiment 1. Error Bars: +/− 1 SE.

required sample size is 24. We lastly enrolled 31 participants (female 22), all undergraduate students from one university, age range 17–24, mean age 20.16, SD = 1.64. Prior to the experiment, participants were instructed by the experimenter with the ERP experiment procedures and operations. Informed consent was obtained from all participants, and they were compensated after the experiment.

3.1.2. Experimental materials

The number of words was increased to ensure the effectiveness of EEG signal averaging. A total of 144 neutral two-character Chinese words was randomly selected from the Chinese Affective Words System (Wang et al., 2008). These words had an emotional valence of 4.93 ± 1.56; and an arousal level of 4.58 ± 1.89. Among them, 84 were to-be-learned words, and 60 were new words for testing.

3.1.3. Procedure

The experiment consisted of two identical sessions of a recognition memory task, differing only in the word sets used. In each session, 42 words were presented during the learning phase, along with 30 novel words introduced in the test phase. A five-minute break separated the two sessions to reduce potential fatigue and minimize carryover effects from the prior session.

In each trial of the learning phase, following the procedure used in Experiment 1, a fixation cross appeared at the center of the screen for 700 ms. After the fixation disappeared, a static word was presented, and participants were instructed to quickly press either the "↑" or "↓" key to start and control the movement of the word. In the consistent condition, the word moved in the same direction as the pressed key; in the inconsistent condition, the word moved in the opposite direction. After the 400 ms movement stopped, the word remained on the screen for an additional 500 ms before disappearing. Participants were then asked to rate their sense of agency (SoA) by responding to the question, "To what extent did you control the movement of the word?". Each condition included 21 trials in the learning phase, with trials from different conditions presented in a randomized order.

In the test phase, a random subset of 20 previously studied words (10 from each condition) was presented together with 30 new words, in random order. Participants were instructed to press the "F" key if they recognized the word as previously learned and the "J" key if they judged it to be new. For words judged as old (i.e., "F" responses), participants were then required to make a Remember/Know judgment: they pressed the "X" key if they could recollect specific details from the learning phase (i.e., "Remember" response) or the "M" key if the word was recognized as familiar without recollection of contextual details ("Know" response). A schematic overview of the procedure is shown in Fig. 3.

3.1.4. Design

For SoA ratings, a single-factor within-subjects design was employed, with the independent variable being action-outcome consistency (consistent vs. inconsistent). For memory task results, a two-factor within-subjects design was employed, with the independent variable being action-outcome consistency (consistent vs. inconsistent) and response (R vs. K). The dependent variables were the recognition performance. The ERP indices focused on the changes in FN400 and LPC.

3.1.5. Electrophysiological ERP recordings and pre-processing

A Neuroscan 32-channel EEG system was utilized, and data were collected using Curry 7.0 software. During the experiment, participants wore a 32-electrode cap, with electrodes placed according to the international 10–20 system. Additionally, four electrodes were used to detect eye movements and blinks: two electrodes were placed above and below the left eye and recorded vertical electrooculogram (VEOG) activity (approximately 1 cm above the eyebrow and 1 cm below the eyelid), and two electrodes were placed at the outer canthi of both eyes and recorded horizontal electrooculogram (HEOG) activity. The impedance between the scalp and electrodes was kept below 5 kΩ.

The ERP data was imported and analyzed using the open-source toolbox EEGLAB running on MATLAB. During preprocessing, the EEG signals were sampled at a frequency of 500 Hz, and filtered with band-pass filter between 0.05 and 40 Hz. The average of the bilateral mastoids served as the reference for analysis. Epochs were extracted from 200 ms before word stimulus onset to 1600 ms after stimulus onset, with the average activity during the −200 to 0 ms interval (pre-stimulus) serving as the baseline, and re-referenced to an average of the bilateral mastoid channels. Independent Component Analysis (ICA) was used to correct for blink and motion artifacts, and components with amplitudes exceeding ±100 μV were excluded as artifacts. The artifact rejection rate was kept below 25%, and the number of correct recognition trials was at least 20 for each condition, for each participant.

By observing waveform plots and referencing previous studies (Hou et al., 2013), the time window for the FN400 component was determined to be 300-500 ms, and the time window for the LPC component was determined to be 500-800 ms. The electrodes of interest included frontal (Fz), frontocentral (FCz), central (Cz), centroparietal (CPz), and parietal (Pz) sites (Leynes and Mok, 2017; Leynes and Crawford, 2018).

3.2. Behavioral results

We calculated the correct R/K judgments for old words, and also R/K judgments for new words in each consistency condition for each participant. The R (and K) performance was calculated by subtracting the R (K) proportion for new words from that of R (K) for old words

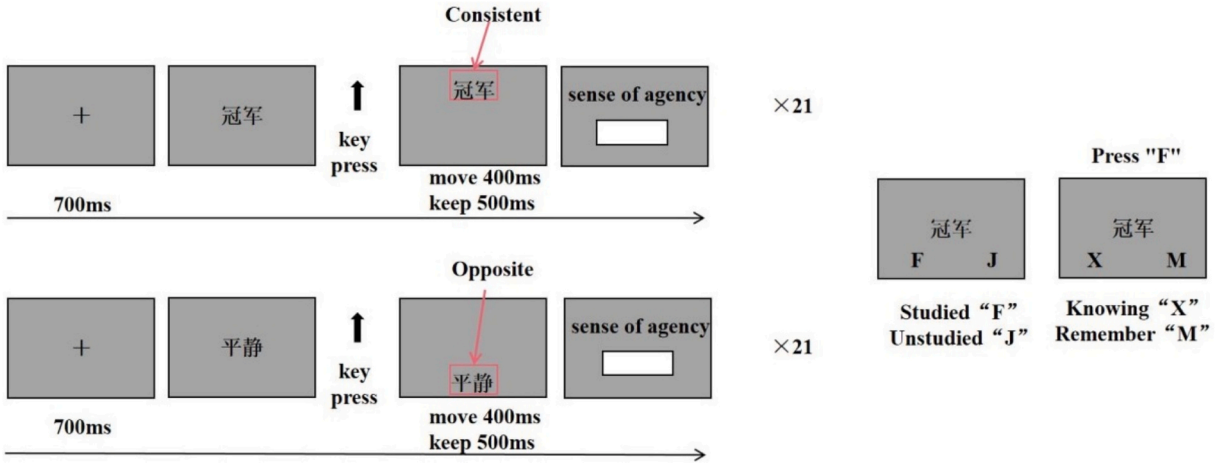


Fig. 3. Experimental procedure of the R/K paradigm in Experiment 2.

studied for all conditions. A higher R/K performance indicates that the participants could better recollect/was familiar with old words and judged new words as new (Rogers et al., 1977; Gardiner, 1988; Tsuji and Imaizumi, 2022).

We used one factor repeated measure ANOVA to analyze the SoA data, with the within factor consistency (consistent or inconsistent). The results yielded a significant main effect of consistency, $F(1,30) = 53.01$, $p < .001$, $\eta p2 = 0.64$, BF > 100, due to significantly higher SoA ratings in the consistent condition (Mean = 5.84, SE = 0.24) than in the inconsistent condition (Mean = 3.09, SE = 0.33).

We then ran 2 (conditions: consistent, inconsistent) * 2 (R and K response) ANOVA analysis on R/K memory performance, which showed that the main effect of consistency was not significant, $p = .713$, BF10 = 0.25, as was the interaction, $p = .672$, BF10 = 0.26. However, the main effect of R/K response was significant, $F(1,30) = 10.54$, $p = .003$, $\eta p2 = 0.26$, BF10 = 13.11, indicating that the R response score (Mean = 0.23, SE = 0.03) was significantly higher than the K response score (Mean = 0.13, SE = 0.02). Please see Fig. 4.

3.3. EEG results

Topographic maps (see Fig. 5) suggested that the FN400 was at max at mid-frontal electrode sites (Leynes and Mok, 2017), so that the analysis was restricted to the mid-frontal electrodes (i.e., F3, Fz, F4, FC3, FCz, and FC4). For FN400, we ran a 3 (conditions: consistent, inconsistent, new word) × 2 (Anterior/Posterior electrode position: front and back) × 3 (Left/Right electrode position: left, middle and right) repeated measures ANOVA, with Greenhouse–Geisser correction applied to all repeated measures when the Mauchley test result is significant.

The results indicated that the main effect of condition just missed significance, $F(2,60) = 3.02$, $p = .056$, $\eta p2 = 0.09$, BF10 = 0.53, while the main effect of LR electrodes position was significant, $F(2,60) = 9.16$, $p < .001$, $\eta p2 = 0.23$, BF10 = 52.5. No other significant effect was found, ps > 0.35. Post-hoc analysis on the three conditions revealed the following: FN400 was significantly higher in consistent (Mean = 0.01, SE = 1.01) than in inconsistent conditions (Mean = −1.95, SE = 0.93), mean difference = 1.97, SE = 0.90, $p = .036$; and numerically higher than new word conditions (Mean = −1.09, SE = 0.74), $p = .15$. But this result should be considered exploratory given the non-significant analysis and the absence of Bayesian support. No other significant difference was found, all $p = .26$. Post-hoc analysis on the three LR positions revealed the following: FN400 was significant lower in the middle (Mean = −1.57, SE = 0.85) than at the left (Mean = −0.48, SE = 0.68), mean difference = 1.09, SE = 0.26, $p < .001$; and right (Mean = −0.97, SE = 0.81), mean difference = 0.60, SE = 0.21, $p = .008$. No significant difference was found between left and right electrodes, $p > .09$.

The ERPs and topographic maps for the LPC suggested a locus at left-parietal electrodes, as in previous studies. Therefore, the analyses were restricted to left-parietal electrodes (i.e., C3, CP3, and P3). We ran a 3 (conditions: consistent, inconsistent, new word) × 3 (electrode position along the midline-parietal axis: front, middle and back) repeated measures ANOVA, with Greenhouse–Geisser correction applied to all repeated measures when the Mauchley test result is significant. The results showed that no effect was significant, ps > 0.18, although LPC was numerically higher in consistent (Mean = 2.47, SE = 1.02) than in inconsistent conditions (Mean = 2.00, SE = 0.67), $p = .103$; and numerically higher in consistent than new word (Mean = 0.88, SE = 0.48), $p = .115$.

3.4. Discussion

The behavioral results confirmed that the manipulation was successful: participants experienced a stronger sense of agency (SoA) in the consistent condition and a weaker SoA in the inconsistent (opposite) condition, indicating that action–outcome consistency reliably modulates SoA (Ebert and Wegner, 2010; Zopf et al., 2018).

With respect to memory performance, however, neither the behavioral data nor the LPC component of the ERP supported our hypothesis. Specifically, no significant differences were observed between the consistent and inconsistent conditions for either Remember (R) or Know (K) responses. These findings suggest that SoA did not influence recognition memory performance under the present conditions, consistent with some previous findings (Tsuji and Imaizumi, 2022) but inconsistent with the results of our Experiment 1 and those of Hon and Yeo (2021).

Nevertheless, the ERP results yielded more nuanced and potentially informative patterns. For the FN400 component, which is typically associated with familiarity-based ("Know") recognition, amplitudes in the consistent condition were marginally larger than those observed in the inconsistent and new-word conditions. Similarly, for the LPC component, which is linked to recollection-based ("Remember") recognition, amplitudes in the consistent condition were numerically greater than those in the inconsistent and new-word conditions. Although these effects did not reach conventional levels of statistical significance, they suggest that action–outcome consistency may modulate both familiarity and recollection processes at the neural level.

We propose two complementary explanations for this pattern of

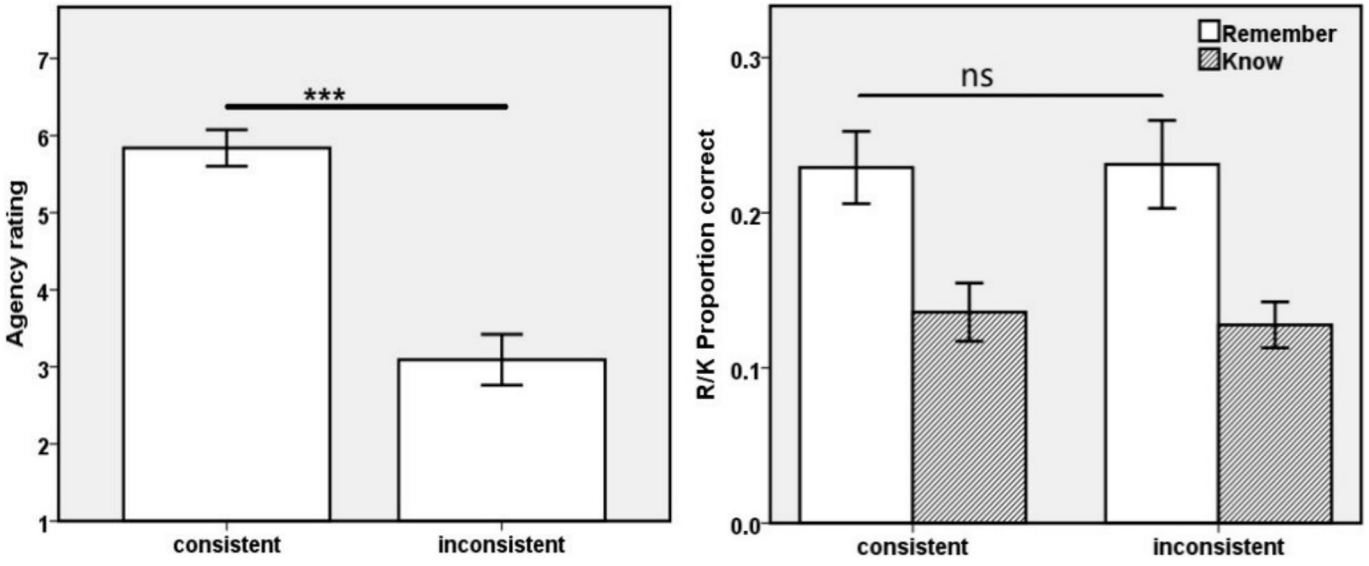
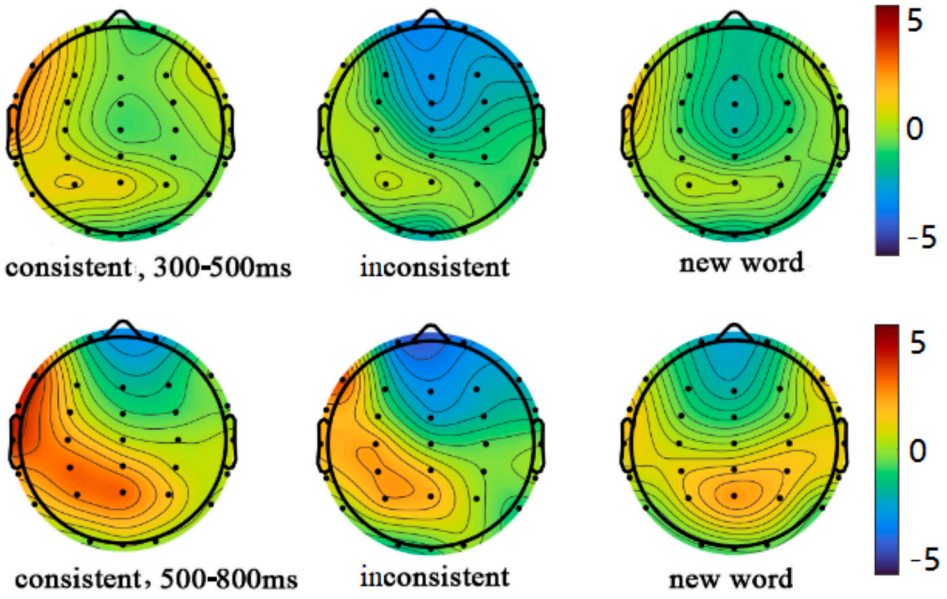


Fig. 4. Agency ratings (left) and R/K memory performance (right). Error Bars: +/− 1 SE.



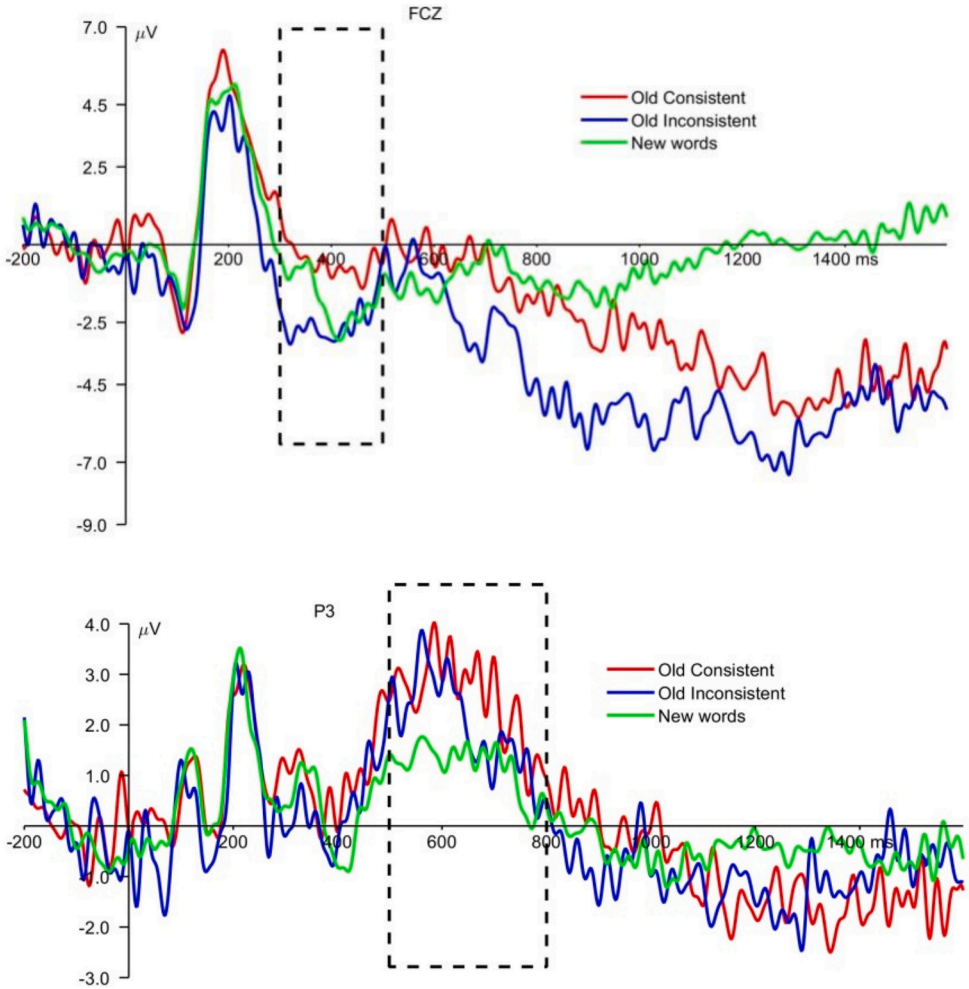


Fig. 5. Grand-average ERPs during the test phase depicting the FN400 (300-500 ms) and LPC (500-800 ms) in Experiment 2. Waveforms are shown from electrode FCZ for FN400, and P3 for LPC. Topographic maps plot differences across the scalp among the three conditions.

results. First, the R/K paradigm requires participants to introspect and categorize their memory experience, which consumes additional executive resources. This added cognitive load may compete with the limited-capacity encoding benefits provided by SoA, particularly when the SoA manipulation is subtle. Consequently, the memory advantage observed in the simple recognition task (Experiment 1) was diluted in the more demanding R/K paradigm. Second, the difference in SoA induced by action–outcome consistency may have been relatively subtle, such that the memory enhancement observed in Experiment 1 was diminished in Experiment 2 when combined with the more demanding R/K procedure.

Accordingly, because the R/K paradigm remains critical for examining the potential role of self-referential processing in SoA-related memory effects, Experiment 3 and 4 was designed to amplify the strength of the SoA manipulation. Specifically, we sought to increase the contrast in perceived SoA between experimental conditions in order to more sensitively detect potential behavioral and neural effects on recognition memory.

4. Experiment 3

According to prominent models of the SoA (Synofzik et al., 2008), action–outcome consistency primarily influences the *feeling of agency*, a relatively low-level, implicit, and non-conceptual component of SoA. In contrast, intentions and beliefs are more closely associated with *agency attribution*, a higher-level, explicit, conceptual, and interpretative component (Wegner, 2002). Given that the manipulation in Experiment 1 relied exclusively on action–outcome consistency, it likely targeted the feeling of agency, resulting in a relatively weak differentiation in overall SoA strength in the two consistency conditions that may have been insufficient to produce detectable effects on memory performance.

In Experiment 3 and 4, we therefore emphasized action volition, based on the assumption that voluntary actions are more strongly self-related and thus more likely to engage self-referential processing. From this perspective, in Experiment 1 and 2 participants always executed voluntary keypresses in all conditions, such that volitional aspects of action were held constant. This lack of variation in action volition may explain why differences in self-referential processing across action–outcome consistency conditions were difficult to detect.

There is direct empirical support for this assumption. First, a substantial body of research has demonstrated that voluntary action plays a critical role in shaping SoA (Caspar et al., 2016; Giersiepen et al., 2023; Haggard et al., 2002; Lepron et al., 2015). For example, participants who initiate actions themselves report a stronger SoA than those who merely follow another agent's actions (Obhi and Hall, 2011). Electrophysiological studies further indicate that coerced or involuntary actions are associated with reduced SoA, as reflected by more negative N1 amplitudes relative to voluntary actions (Caspar et al., 2016). This SoA reduction has been attributed to diminished feelings of responsibility for action outcomes under involuntary conditions (Frith, 2014; Caspar et al., 2020).

Second, prior research indeed has shown that voluntary actions enhance memory relative to involuntary actions (Houser et al., 2022; Murty et al., 2015). For instance, Murty et al. (2015) allowed participants either to freely choose between left and right keypresses or to follow externally imposed prompts. Their results demonstrated superior declarative memory performance in the free-choice condition, indicating that volitional control during encoding enhances memory, even when the choice itself is unrelated to the to-be-remembered material.

Third, in Experiment 3 and 4 the action-outcome consistency was held constant across conditions, thereby ruling out the confounding influence of prediction-error-related attentional capture that was possibly present in Experiments 1 and 2.

Taken together, these findings imply that voluntary action may more effectively amplify the influence of SoA on recognition memory than action–outcome consistency alone, particularly under the cognitively demanding R/K paradigm. Accordingly, in Experiment 3 and 4 we manipulated SoA by varying action volition (voluntary vs. involuntary actions). Specifically, in Experiment 3 we only manipulated volition and aimed to validate that SoA, induced through voluntary action, can also facilitate general recognition memory performance; while in Experiment 4 we retained the same behavioral R/K paradigm and electrophysiological measures used in Experiment 2. This design allowed us to further investigate the effects of SoA on recognition memory and to examine the potential mediating role of self-referential processing.

4.1. Method

4.1.1. Participants

Similar to Experiment 1, we performed an a-priori power analysis with G power (Faul et al., 2007), with a medium effect size (f = 0.25), α = 0.05, power = 0.8, number of measurements = 2, and found that the required sample size is 34. A total of 40 college students (female 22) from one university, age range 18–25, mean age 21.22, SD = 1.90, were recruited, to ensure sufficient statistical power.

4.1.2. Experimental Materials

From the Chinese Affective Word System (Wang et al., 2008), a total of 216 neutral two-character words were randomly selected as the experimental keywords. These words had an emotional valence of 4.94 ± 1.57 and an arousal level of 4.59 ± 1.90. Among them, 126 were to-be-learned words, and 90 new words for testing.

4.1.3. Procedure

First, a fixation point appeared at the center of the screen for 700 ms. Then, in the voluntary condition, a word appeared, and participants could freely choose to press the "↑" or "↓" key to control the movement of the word. In the involuntary condition, a cue appeared following the fixation point, and participants were instructed to press the key indicated by the cue. After the key press, the word moved for 400 ms to the top or bottom of the screen, and then stopped for 500 ms. The movement of the word was always consistent with the pressed key in both voluntary and involuntary conditions. The remaining procedures were the same as in Experiment 2. All used words were randomly divided into the three sessions and presented (42 × 3, 30 × 3) to participants. There were five minute breaks between them. Please see Fig. 6.

4.1.4. Design

We used a single-factor within-participants design for both SoA and memory task, with the independent variable condition (voluntary action vs. involuntary action). In the voluntary action condition, participants freely choose which key "↑" or "↓" to press, and then the word moved consistently; in the involuntary action condition, participants followed the instructions and press one key "↑" or "↓", then the word always moved consistently. The dependent variables were consistent with Experiment 1.

4.2. Behavioral Results

We used one factor repeated measure ANOVA to analyze the SoA data, with the within factor volition (voluntary or involuntary). The results indicated a significant main effect of consistency, $F(1,39) = 139.60$, $p < .001$, $\eta p2 = 0.78$, BF > 100, showing that SoA ratings were significantly higher in the voluntary condition (Mean = 6.63, SE = 0.09) than in the involuntary condition (Mean = 4.91, SE = 0.13).

We calculated the memory performance as in Experiment 1. In the recognition memory task, one factor repeated measure ANOVA results indicated a significant main effect of volition, $F(1,39) = 14.13$. $p = .001$, $\eta p2 = 0.27$, showing that recognition performance scores significantly higher in the voluntary action condition (Mean = 0.57, SE = 0.03) than in the involuntary action condition (Mean = 0.47, SE = 0.02). Bayesian analysis revealed that BF10 = 51.30, providing very strong evidence for

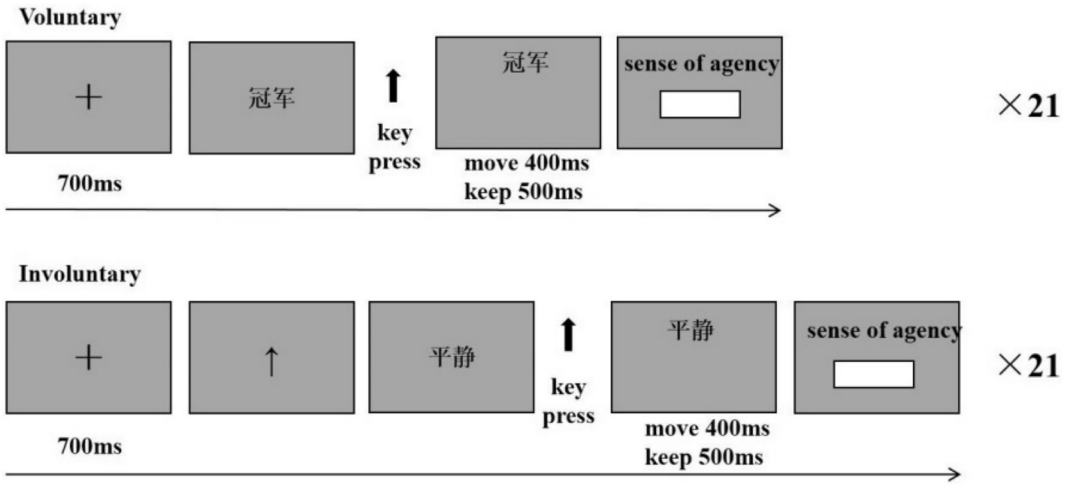


Fig. 6. Experimental procedure of the R/K paradigm for the Experiment 3.

our hypothesis. Please see Fig. 7.

4.3. Discussion

In the Experiment 3, the SoA ratings demonstrated that participants experienced a stronger SoA in the voluntary condition than in the involuntary condition, as in previous research (Caspar et al., 2016; Haggard et al., 2002). Importantly, recognition memory was also significantly higher in the voluntary condition than in the involuntary condition. Accordingly, we successfully validated that SoA, manipulated through voluntary action rather than action-outcome consistency, can also improve recognition memory (Houser et al., 2022; Murty et al., 2015). Notably, the recognition memory performance difference (mean = 0.09, SE = 0.02) in two voluntary conditions in Experiment 3 seemed numerical larger than the difference (mean = 0.04, SE = 0.01) in two consistency conditions in Experiment 1. This pattern is consistent with the possibility that volition-based manipulation may produce a stronger SoA contrast than consistency-based manipulation, thereby enhancing the detectability of memory effects and self-referential processing.

5. Experiment 4

As discussed above, and based on the findings from the three experiments, given that we assume that the possibility reason of the null-effect reported by Tsuji and Imaizumi (2022) might be related to the use of a demanding paradigm (R/K); and the SoA strength difference induced through the manipulation of volition (voluntary vs. involuntary) was stronger than through the manipulation of consistency (consistent vs. inconsistent), which was used in Tsuji and Imaizumi (2022); we may infer that, if we combine the volition manipulation with the R/K paradigm, we may expect more detectable effects. Thus, in Experiment 4, we continue to investigate the role of the research paradigm and its self-referential implications for the memory enhancement effect, through the manipulation of volition. We hypothesized that, if SoA enhances memory via self-referential mechanisms, voluntary conditions would yield higher "Remember" performance and larger LPC amplitudes than involuntary and new-word conditions; while voluntary and involuntary conditions may yield no difference in either the "Know" performance and FN400 amplitude.

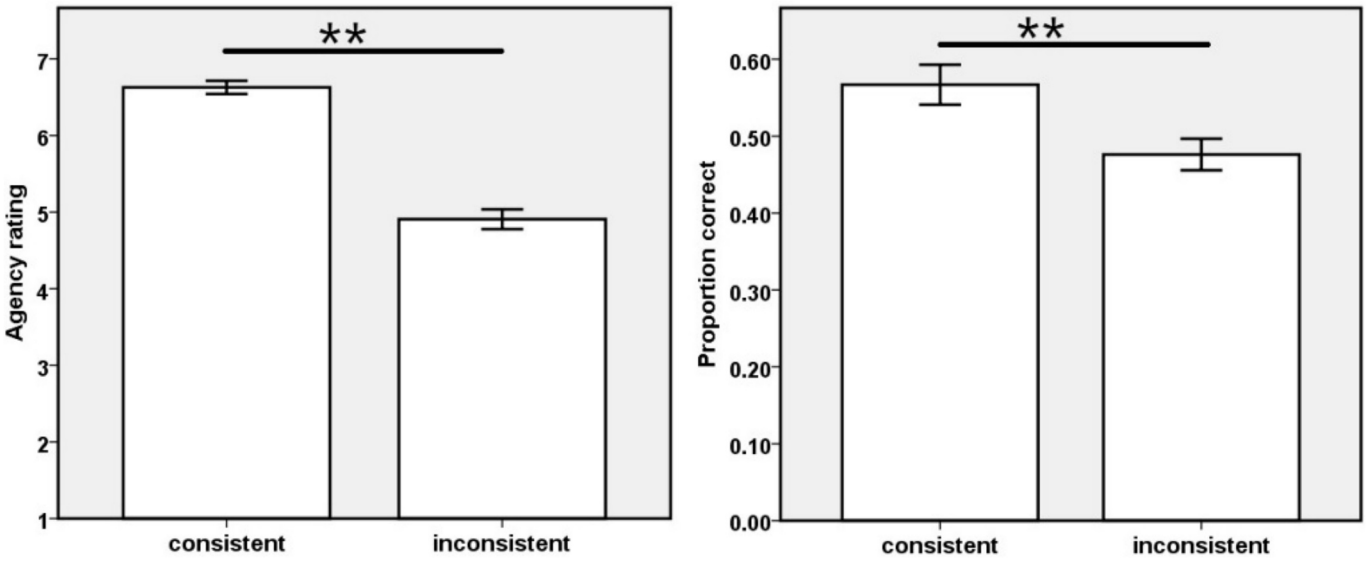


Fig. 7. Agency ratings (left) and recognition memory performance (right). Error Bars: +/− 1 SE.

5.1. Method

5.1.1. Participants

Similar to Experiment 2, we performed an a-priori power analysis with G power (Faul et al., 2007), with a medium effect size (f = 0.25), α = 0.05, power = 0.8, number of measurements = 4, and found that the required sample size is 24. A total of 29 college students (female 19) from one university, age range 17–23, mean age 19.83, SD = 1.54, were recruited.

5.1.2. Experimental Materials and Procedure

The Experimental Materials, i.e., the used words, were the same as used in Experiment 3. The procedure was a combination of Experiment 3 and 2. First, a fixation point appeared at the center of the screen for 700 ms. Then, in the voluntary condition, a word appeared, and participants could freely choose to press the "↑" or "↓" key to control the movement of the word. In the involuntary condition, a cue appeared following the fixation point, and participants were instructed to press the key indicated by the cue. After the key press, the word moved for 400 ms to the top or bottom of the screen, and then stopped for 500 ms. The movement of the word was always consistent with the pressed key in both voluntary and involuntary conditions. That is, the learning phase was the same as in Experiment 3. And the test phase was the same as in Experiment 2, participants needed to make a old/new judgment and then Remember/Know judgment if the current word was judged as old: they pressed the "X" key if they could recollect specific details from the learning phase (i.e., "Remember" response) or the "M" key if the word was recognized as familiar without recollection of contextual details ("Know" response). The 126 to-be-learned words were divided into three equal lists of 42 words, each serving as the study set for one of the three experimental sessions. Similarly, the 90 new words were divided into three lists of 30 words for the test phases. The three sessions were separated by 5-minute breaks, and the order of word lists was counterbalanced across participants. Please see Fig. 8.

5.1.3. Design

For SoA ratings, we used a single-factor within-participants design, with the independent variable condition (voluntary vs. involuntary action) as used in Experiment 3. The dependent variables were consistent with Experiment 2. For memory task results, a two-factor within-subjects design was employed, with the independent variable being action volition (voluntary vs. involuntary) and response (R vs. K). The dependent variables were the recognition performance. The ERP indices focused on the changes in FN400 and LPC.

5.2. Behavioral results

We used one factor repeated measure ANOVA to analyze the SoA data, with the within factor consistency (consistent or inconsistent). The results indicated a significant main effect of consistency, $F(1,28) = 55.63$, $p < .001$, $\eta p2 = 0.67$, BF10 > 100, showing that SoA ratings were significantly higher in the voluntary condition (Mean = 6.74, SE = 0.07) than in the involuntary condition (Mean = 4.88, SE = 0.24).

A 2 (conditions: voluntary, involuntary) * 2 (R and K response) repeated measures ANOVA of R/K memory performance yielded a significant main effect of voluntary condition, $F(1, 28) = 5.41$, $p = .028$, $\eta p2 = 0.16$, with responses being more accurate in voluntary (Mean = 0.29, SE = 0.02) than involuntary conditions (Mean = 0.23, SE = 0.02), however the corresponding Bayes factor (BF10 = 0.71) indicated only anecdotal evidence, did not provide compelling evidence in favor of the alternative hypothesis. According to our pre-specified interpretation criteria, the hypothesized effect is inconclusive. The main effect of R/K response was also significant, $F(1, 28) = 37.28$, $p < .001$, $\eta p2 = 0.57$, BF10 > 100, with R responses (Mean = 0.40, SE = 0.03) being significantly more accurate than K responses (Mean = 0.12, SE = 0.03). Critically, the interaction between voluntary action and R/K responses was also significant, $F(1, 28) = 4.91$, $p = .035$, $\eta p2 = 0.15$, BF10 = 11.76, providing strong evidence for the interaction.

We continued to analyze the simple effect, used the paired *t*-test, and found that, the volition difference was significant for R response, $t(28) = 2.70$, $p = .012$, $d = 0.50$; but not for K response, $p = .42$. And the R/K difference was significant for both voluntary, $t(28) = 5.21$, $p < .001$, $d = 0.97$; and involuntary action conditions, $t(28) = 4.65$, $p < .001$, $d = 0.86$. See Fig. 9.

5.3. EEG results

As in Experiment 2, the FN400 analysis was restricted to the midfrontal electrodes (i.e., F3, Fz, F4, FC3, FCz, and FC4). For FN400, the 3 (conditions: voluntary, involuntary, new word) × 2 (Anterior/Posterior electrode position: front and back) × 3 (Left/Right electrode position: left, middle and right) repeated measures ANOVA indicated that the main effect of condition was significant, $F(2,56) = 4.25$, $p = .019$, $\eta p2 = 0.13$, BF10 = 4.84. There was no other significant effect, ps > 0.13. Post-hoc analysis on the three conditions revealed that: FN400 in voluntary (Mean = 1.28, SE = 0.63) was significant higher than involuntary (Mean = −0.32, SE = 0.82), mean difference = 1.59, SE = 0.67, $p = .024$; and higher in voluntary than new words (Mean = −0.04, SE = 0.50), mean difference = 1.32, SE = 0.49, $p = .012$. No difference between involuntary and new words, $p = .644$.

As in Experiment 2, the LPC analyses were restricted to left-parietal

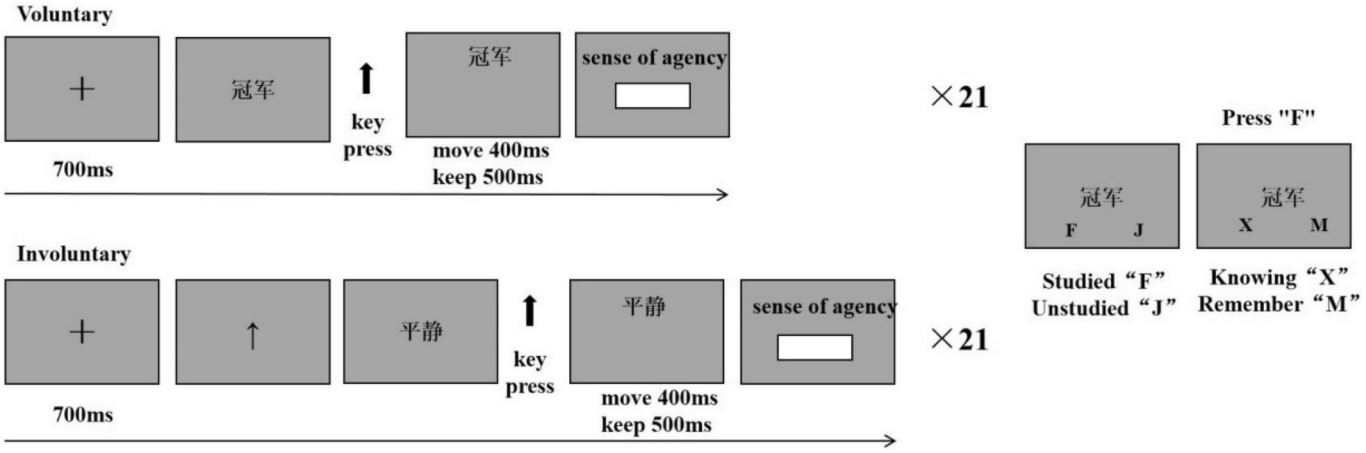


Fig. 8. Experimental procedure of the R/K paradigm for the Experiment 4.

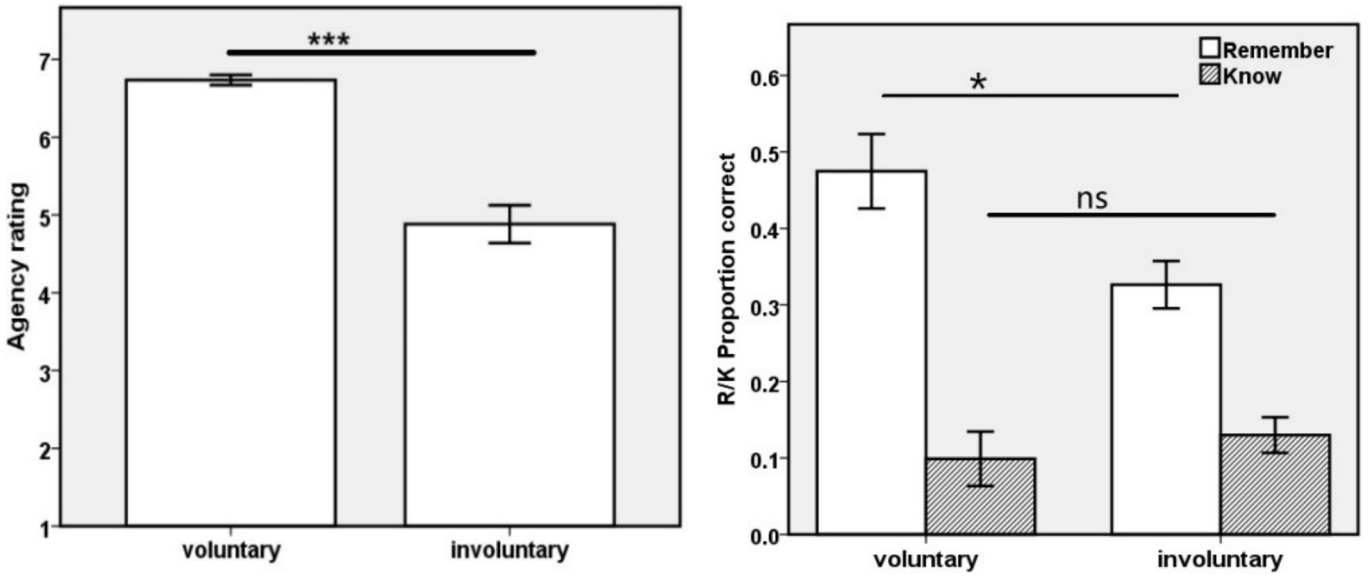


Fig. 9. Agency ratings (left) and R/K memory performance (right). Error Bars: +/− 1 SE.

electrodes (i.e., C3, CP3, and P3). For LPC, a 3 (conditions: voluntary, involuntary, new word) × 3 (electrode position along the midline-parietal axis: front, middle and back) repeated measures ANOVA indicated a significant main effect of voluntary condition, $F(2,56) = 4.73$, $p = .013$, $\eta p2 = 0.14$, BF10 = 6.36. There was no other significant effect, ps > 0.50. Post-hoc analysis on the three conditions revealed that: LPC in voluntary (Mean = 3.63, SE = 0.82) was significant higher than involuntary (Mean = 2.05, SE = 0.65), mean difference = 1.57, SE = 0.58, $p = .011$; and higher in voluntary than new words (Mean = 1.84, SE = 0.55), mean difference = 1.79, SE = 0.80, $p = .034$. No difference between involuntary and new words, $p = .658$. See Fig. 10.

5.4. Discussion

The Experiment 2 successfully manipulated the SoA, with participants experiencing a stronger SoA under voluntary than under involuntary conditions. This aligns with previous research and indicates that voluntary action significantly influences the perceived SoA (Tsakiris and Haggard, 2003; Wenke et al., 2009).

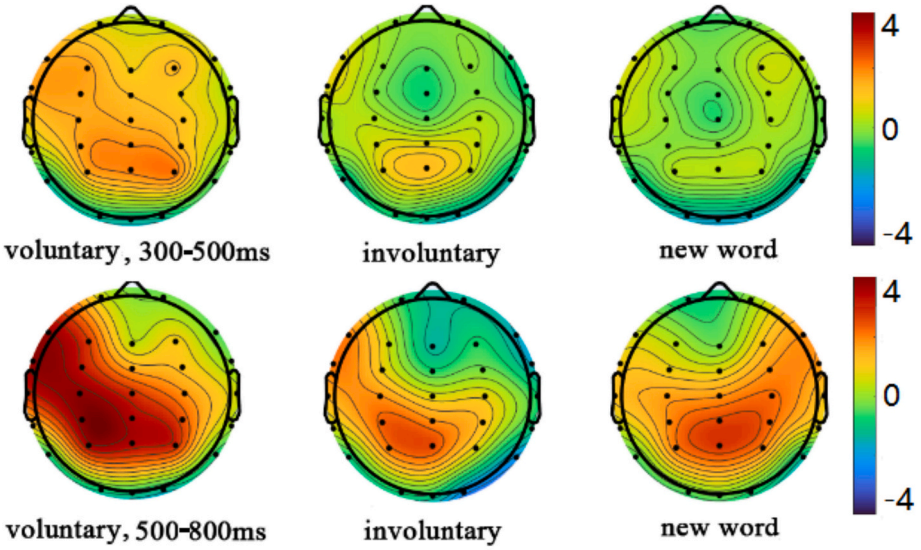
Importantly, our behavioral findings partially support the self-referential hypothesis, as voluntary action selectively enhanced Remember responses without affecting Know responses. However, the ERP results revealed a more complex pattern: voluntary conditions enhanced both FN400 (familiarity) and LPC (recollection) amplitudes. This suggests that volitional control may boost overall memory strength rather than selectively enhancing recollection, the behavioral R/K separation emerges from differential decision criteria applied to a continuous memory signal, rather than from qualitatively distinct memory systems.

6. General discussion

Previous research has proposed self-referential processing as a key mechanism through which the sense of agency (SoA) influences memory (Hon and Yeo, 2021; Tsuji and Imaizumi, 2022). Specifically, experiencing SoA over stimuli is assumed to increase their self-relatedness, thereby enhancing encoding and retrieval and ultimately improving memory performance (Hon and Yeo, 2021). However, Tsuji and Imaizumi (2022), using the remember/know (R/K) paradigm, failed to find evidence supporting a self-referential account. In the present study, we addressed this discrepancy through four experiments designed to provide a more comprehensive investigation. In Experiment 1, we confirmed that SoA induced via action–outcome consistency enhances recognition memory; while the difference between two consistency conditions was small. In the Experiment 3, we confirmed that SoA induced via voluntary action also enhances recognition memory; especially the difference between two voluntary conditions was large. In Experiments 2 and 4, we further examined this effect using R/K paradigm and electrophysiological measures. Notably, even Experiment 2 did not reveal any influence from consistency on behavioral and ERP results; Experiment 4, which manipulated action volition, indeed yielded clear behavioral and ERP evidence supporting a self-referential mechanism underlying the memory enhancement associated with SoA.

According to the comparator model and the two-step account of SoA (Synofzik et al., 2008; Haggard, 2017; Moore, 2016; Yoshie & Haggard, 2017), action–outcome consistency and voluntary action reflect distinct components of agency. Action–outcome consistency primarily contributes to the non-conceptual feeling of agency, which is based on sensorimotor matching between predicted and actual outcomes. In contrast, voluntary action more strongly engages explicit agency attribution, which relies on higher-level conceptual processes, intentions, and beliefs in conjunction with agency feelings. Applied to the present experiments, in Experiment 1 and 2, participants freely executed keypresses in all conditions, such that differences in self-referential information across action–outcome consistency conditions were relatively subtle. Combined with the increased cognitive demands imposed by the R/K paradigm in Experiment 2, this likely attenuated observable effect on memory recollection performance. In Experiment 3 and 4, however, action volition was directly manipulated: participants either acted voluntarily or followed external instructions. This manipulation produced more pronounced differences in self-related processing, enabling us to detect reliable effects on SoA, R/K performance, and ERP components.

Within the memory literature on the self-reference effect, a widely accepted framework posits that recollection and familiarity are dissociable processes associated with distinct ERP markers. The late positive component (LPC; 500–800 ms), typically observed over parietal regions, is considered an index of conscious, detail-rich recollection, whereas the fronto-central FN400 component (300–500 ms) is associated with familiarity-based recognition (Curran, 2000; Rugg and Curran, 2007). Self-referential encoding has been shown to preferentially enhance especially the LPC, consistent with the view that self-reference strengthens recollection through deeper, more elaborative, and context-rich encoding (Curran and Cleary, 2003; Addante et al., 2012).



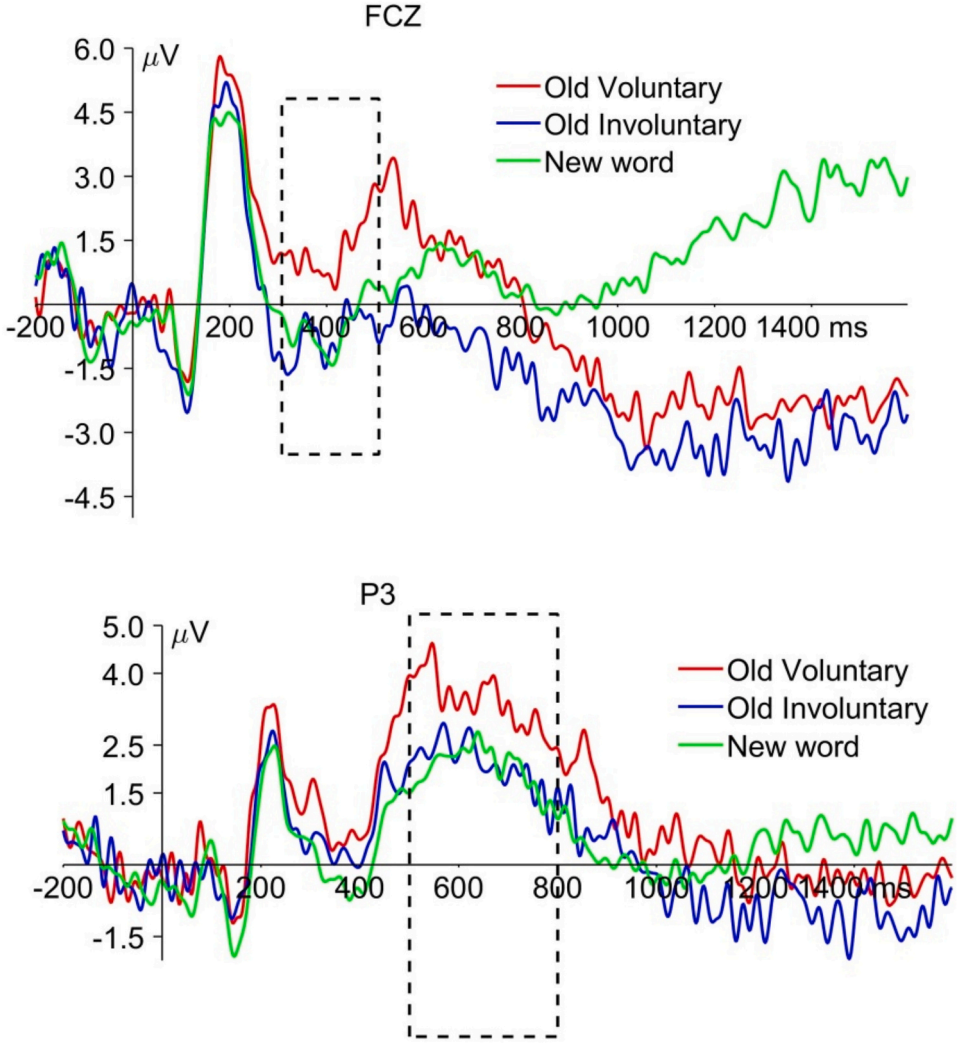


Fig. 10. Grand-average ERPs during the test phase depicting the FN400 (300-500 ms) and LPC (500-800 ms) in Experiment 4. Waveforms are shown from electrode FCZ for FN400, and P3 for LPC. Topographic maps plot differences across the scalp among the three conditions.

Nevertheless, accumulating evidence suggests that recollection and familiarity may not be entirely independent processes, but instead may share overlapping cognitive and neural resources (Skinner and Fernandes, 2007; Herzmann et al., 2011). For example, emotional interference has been shown to impair both recollection and familiarity (Mao et al., 2015). Moreover, self-referential words elicit enhanced old/new effects in both the FN400 and LPC time windows (Pereira et al., 2021), and similar dual-component enhancements have been reported using directed-forgetting paradigms (Mao et al., 2017). These findings suggest that self-referential processing may broadly enhance encoding quality, thereby supporting both familiarity- and recollection-based memory judgments. From this perspective, self-reference may function as a powerful encoding modulator that simultaneously influences multiple dimensions of memory formation. Future research will be needed to determine whether self-referential processing enhances familiarity and recollection in parallel through strengthened self-concept integration, or whether it primarily targets recollection, with familiarity effects emerging as secondary consequences.

It is very interesting to look into our findings in Experiment 4, because behavioral data showed an increase in Remember (R) responses with no change in Know (K) responses, while ERP data showed enhanced FN400 and LPC old/new effects, this results pattern seems contradict to the claims in previous self-referential processing studies: the LPC components should show significant old/new effects because it corresponds to the Remember responses; while FN400 should not because it corresponds to the Know responses. We may argue that the findings provide compelling support for the single-process (memory strength/signal detection) account (Dunn, 2004, 2008; Rotello and Macmillan, 2006), rather than the dual-process account for the Remember/Know task. Specifically, the Remember and Know responses may reflect different levels of a single memory strength dimension, but not two qualitatively distinct processes underlying recognition memory.

The single-process framework, rooted in signal detection theory, conceptualizes memory as a continuous dimension of strength. R and K judgments reflect different decision criteria along this continuum: K corresponds to a lower threshold for familiarity, while R corresponds to a higher threshold for recollection (Wixted and Mickes, 2010; Brezis et al., 2017). If the experimental manipulation simply improves the entire memory strength, the consequence is precisely what we observed: the volition manipulation improved the entire memory strength, the number of items exceeding the high R threshold increases significantly, while the count at the moderate K threshold remains relatively stable, because items moving from K to R are replenished by weaker items crossing the K boundary from below.

Crucially, our ERP data provide the neural signature of this strength enhancement. The FN400, typically linked to familiarity, and the LPC, typically linked to recollection, both increased under the manipulation. It seems that the FN400 and LPC may not serve as "process-pure" markers of familiarity and recollection in our study. Within the single-process view, both components are sensitive to the same continuous memory signal—they do not index qualitatively distinct systems but rather reflect different temporal stages of retrieval processing. FN400 enhancement may indicate stronger conceptual fluency and mnemonic activation for all studied items (Bader et al., 2023), while LPC enhancement may reflect greater post-retrieval evaluation and confidence (Sun et al., 2024). Their co-variation is exactly what would be expected if the manipulation boosted overall memory strength. In contrast, the dual-process framework would predict a selective effect: if only recollection (R) is enhanced, only the LPC amplitude should be modulated; if familiarity (K) is unaffected, FN400 amplitude should remain unchanged, but this hypothesis was contrary to our observation.

Also, Fröber and Pastötter (2025) provided direct evidence supporting a single-process/signal detection account of free choice on recognition memory. Using the unequal-variance signal detection (UVSD) model, the authors found that free choice selectively enhanced memory strength (d') for relevant (to-be-remembered) items but not for irrelevant distractors. Critically, this selective memory benefit can be parsimoniously explained within a single-process framework: free choice deepens encoding quality, thereby increasing the overall memory strength signal for relevant items. In contrast, a dual-process account, which posits qualitatively distinct recollection and familiarity processes, would require additional assumptions to explain why free choice did not produce a general memory enhancement across all relevant and irrelevant items in their results.

Therefore, our findings align more closely with the single-process/memory strength account than with a strict dual-process model. The null effect in Experiment 2, combined with the significant effect in Experiment 4, further suggests that the detectability of SoA effects depends critically on the strength of the mnemonic signal relative to the decision criteria. When the SoA manipulation is subtle (action-outcome consistency), the resultant strength shift may be insufficient to overcome the cognitive demands of the R/K paradigm, leading to null effects. However, when the manipulation is robust (action volition), the strength shift becomes large enough to produce measurable behavioral and neural effects. This interpretation reconciles the apparent discrepancies in the literature and underscores the theoretical utility of signal detection approaches to understanding recognition memory (Dunn, 2004, 2008; Brezis et al., 2017).

Note that our findings are also broadly consistent with recent neuroscientific evidence demonstrating that the opportunity to make choices facilitates declarative memory. Murty et al. (2015) showed that simply having the opportunity to choose when encoding led to robust enhancements in memory, this effect were revealed to be associated with anticipatory striatal activity and its interaction with the hippocampus (Murty et al., 2019). Similarly, Cheng et al. (2023) found that choice during encoding improved recognition performance after a 24-h delay and that theta oscillations played a significant role in choice-related memory formation. While these studies focused on the opportunity to choose as a source of SoA, our findings extend this line of research by dissociating the specific memory processes—recollection versus familiarity, and revealed interesting behavioral and ERP results.

In addition to self-referential processing, prior research has proposed attentional and reward-based mechanisms to explain how SoA enhances memory (Houser et al., 2022; Zou et al., 2023). One account emphasizes attentional prioritization. By manipulating action–outcome consistency across stimuli, Zou et al. (2023) demonstrated that stimuli associated with stronger SoA attract greater attentional resources, thereby enhancing memory for those stimuli. According to this view, SoA biases attentional selection toward self-controlled information, facilitating encoding (Nakashima, 2019; Wen and Haggard, 2018). However, this explanation is less applicable to the present study, as stimuli were presented sequentially and participants attended to only one stimulus at a time, minimizing competition for attentional priority.

Another account highlights the role of reward. Houser et al. (2022) showed that higher SoA during an interactive task improved memory for event sequences, potentially reflecting engagement of cortico-striatal reward circuitry, particularly the ventral striatum. Prior studies have demonstrated that the opportunity to choose freely can itself function as an intrinsic reward (Leotti and Delgado, 2011), and that both extrinsic rewards and intrinsic motivation enhance memory by modulating expectations and dopaminergic signaling (Gruber and Ranganath, 2019; Murty and Adcock, 2017). Similarly, DuBrow et al. (2019) showed that choice enhances memory through value-related processes. Participants chose or were instructed which occluder screen to remove before viewing a to-be-remembered object. Their results about individual preference and memory enhancement suggested the role of intrinsic value of choice and reward-related mechanisms (Murty et al., 2019). Reward signals may also indirectly influence memory by shaping attentional allocation (Karsh et al., 2016). Thus, in the present experiments, the attentional and reward-based account, i.e., greater volition elicits stronger attention and reward, might predict uniform enhancements in both R and K performance under voluntary conditions. This

prediction is somewhat inconsistent with our observed dissociation between recollection "Remember" and familiarity "Know", suggesting that attention and reward alone cannot fully account for the findings. Further research will be required to clarify the contribution of attentional- and reward-related mechanisms.

In summary, using a combination of behavioral and electrophysiological measures, the present study provides converging evidence that SoA enhances memory through self-referential processing. By distinguishing between different components of agency and directly manipulating action volition, we offer both behavioral and neural support for a self-referential interpretation of SoA-related memory enhancement, thereby extending and refining previous findings in this domain.

CRediT authorship contribution statement

Ruiyu Zhu: Writing – original draft, Methodology, Data curation. **Haorui Ma:** Software, Data curation. **Bernhard Hommel:** Writing – review & editing, Supervision, Project administration. **Ke Ma:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Ethics declaration

Informed consent and patient details

Written informed consent to take part in the study and to publish the article has been obtained from all participants or their legal representatives. The privacy rights of participants have been observed.

Studies in Human

This study was performed in compliance with relevant laws, regulatory frameworks and guidelines where the research took place.

This study was approved by the Southwest university in China. (Approval No. H25040)

Declaration of Generative AI and AI-assisted technologies in the writing process

No generative AI or AI-assisted technologies were used in the writing process for this manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests.

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Ethical approval and informed consent statements

This study was approved by the Ethics Committee of the local university.

Data availability

All raw and processed data supporting this research are publicly available: (https://osf.io/n9s6j/overview?view_only=8074392638d14ad98ab43072b6320482).

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